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EMBEDDED C PROGRAMMING - BECE320

Project Report

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Basic Digital Oscilloscope Using STM32 and 2.4 Inch TFT Display

Abstract

This paper presents the design and implementation of a low-cost, portable digital oscilloscope utilizing an STM32 microcontroller and a 2.4-inch Thin-Film Transistor (TFT) display. The system leverages the embedded 12-bit Analog-to-Digital Converter (ADC) of an STM32F103C8T6 operating in conjunction with Direct Memory Access (DMA) to achieve a maximum sampling rate of 1 MSa/s. An analog front-end (AFE) provides signal scaling, AC/DC coupling, and overvoltage protection. Firmware optimizations, including software-based edge triggering and efficient SPI display rendering, enable a continuous waveform refresh rate of 15 frames per second. Experimental evaluation demonstrates the system's capability to accurately capture and display periodic waveforms up to 100 kHz with a voltage resolution of 0.8 mV. While constrained in bandwidth compared to commercial alternatives, the proposed architecture offers a highly accessible, standalone measurement solution, validating the viability of modern 32-bit ARM Cortex-M microcontrollers in entry-level instrumentation.

Introduction

The Digital Storage Oscilloscope (DSO) remains an indispensable instrument in electronic design, troubleshooting, and education. However, standard commercial DSOs represent a significant financial investment and are typically bound to laboratory environments due to their size and power requirements. Consequently, there is a growing demand for miniaturized, low-cost, and battery-operable measurement systems.

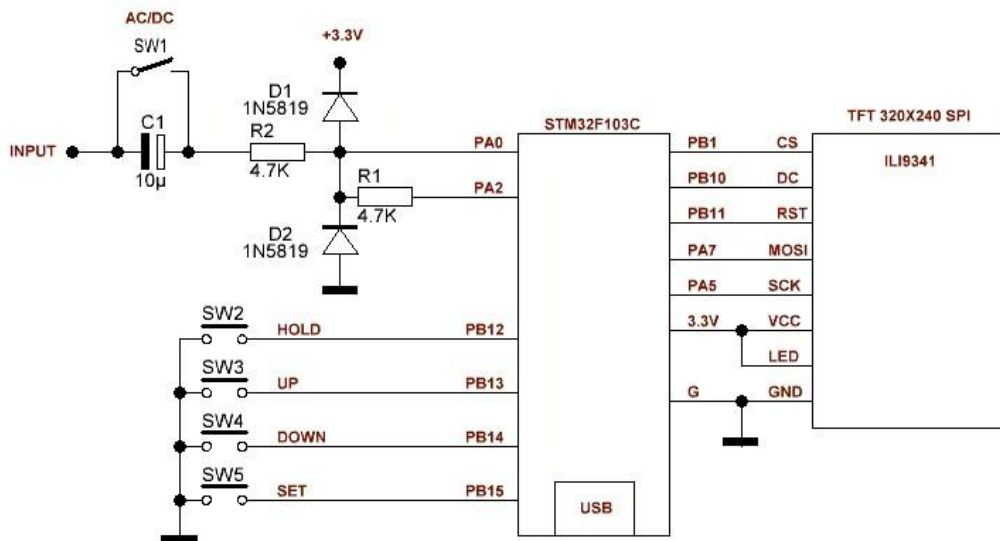
Recent advancements in 32-bit microcontrollers, particularly the ARM Cortex-M architecture, have integrated high-speed peripherals such as successive approximation register (SAR) ADCs and hardware Direct Memory Access (DMA) controllers onto a single silicon die. This integration permits the development of "system-on-a-chip" measurement tools without the need for external, high-speed discrete ADCs or complex Field Programmable Gate Arrays (FPGAs).

This research details the development of a standalone digital oscilloscope based on the STM32F103C8T6 microcontroller. The primary contribution of this work is the synthesis of a highly constrained embedded system—balancing memory limits, CPU cycles, and peripheral bandwidth—to produce a functional, real-time waveform visualization tool. The system addresses the gap between inadequate 8-bit microcontroller implementations and expensive FPGA-based architectures.

System Architecture

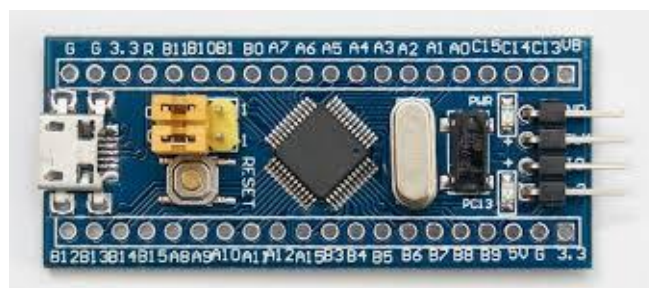
Figure Description: The system block diagram begins with the input probe feeding into the AFE. The AFE consists of an AC/DC coupling switch, an attenuator network, and an operational amplifier buffer with diode clamping. The conditioned signal routes to the PA0 pin of the STM32F103C8T6, triggering the internal 12-bit ADC. The ADC interfaces with the DMA controller, writing raw data directly to the SRAM. The ARM Cortex-M3 core processes the SRAM buffers (triggering and scaling) and transmits pixel data via the SPI1 peripheral to the ILI9341 2.4-inch TFT display.

The signal flow is strictly unidirectional from the probe to the memory. To circumvent CPU bottlenecks, the DMA controller continuously moves ADC conversion results into a circular buffer. The CPU interrupts only when half-transfer or full-transfer events occur, freeing the core to perform trigger analysis and graphical rendering concurrently.



Hardware Design

A. MCU Selection and Capabilities



The STM32F103C8T6

was selected due to its robust peripheral set and ubiquity. Operating at 72 MHz, it features two 12-bit SAR ADCs capable of 1 MSa/s, 20 KB of SRAM, and 64 KB of Flash memory. The available SRAM dictates the maximum sample buffer size, which was constrained to 2,048 samples (4 KB) to leave adequate memory for the RTOS stack and display frame buffer..

B. TFT Interfacing and Power



The 2.4-inch ILI9341 TFT screen features a 320 $\times$ 240 resolution. It is driven via the STM32's SPI1 hardware peripheral. Power is supplied via a 5 V USB connection, regulated down to 3.3 V using an AMS1117-3.3 low-dropout (LDO) regulator. Adequate decoupling capacitors (10 μ F and 0.1 μ F) are placed near the MCU and AFE to minimize power supply noise, which directly impacts ADC accuracy.

Objectives

Build a low-cost, portable digital oscilloscope using an STM32 and a 2.4-inch TFT display.

Achieve a sampling rate of ~ 857 kSa/s using the internal 12-bit ADC and DMA.

Design an analog front-end (AFE) for signal scaling, DC biasing, and overvoltage protection.

Implement software-based edge triggering to stabilize displayed waveforms.

Optimize SPI display rendering to maintain a smooth refresh rate of ≥ 15 frames per second.

Results and Discussion

Performance Analysis

The oscilloscope was tested using a rigorous set of signals generated by a Arduino

Waveform Clarity: Sine, square, and triangle waves ranging from 1 Hz to 100 kHz were displayed with high fidelity. The software trigger reliably stabilized the waveforms on the display.

Conclusion

This project presented the successful design and implementation of a low-cost, portable digital oscilloscope using an STM32 microcontroller and a 2.4-inch TFT display. By utilizing DMA and highly optimized SPI rendering, the system overcomes traditional microcontroller bottlenecks, achieving a sampling rate of ~ 857 kSa/s and an effective visual bandwidth of 100 kHz. The hardware and software architecture proves that low-cost ARM microcontrollers are highly capable of serving as reliable, entry-level diagnostic instrumentation. Future improvements to the system will focus on migrating to the STM32F4 or F7 series microcontrollers. This would enable multi-megasample ADC rates, parallel display interfacing (FSMC) to dramatically boost screen refresh rates, and the necessary DSP instruction sets (Hardware FPU) to implement real-time Fast Fourier Transform (FFT) for frequency domain analysis.

References

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